

# Wei Chen

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## Education

### University of Wisconsin–Madison

Aug 2024 – Dec 2025

M.S. in Electrical and Computer Engineering

Machine Learning and Signal Processing (MLSP), GPA: 3.73 / 4.00

### Sichuan University

Sep 2020 – Jun 2024

B.S. in Flight Vehicle Control and Information Engineering

GPA: 3.79 / 4.00

## Research Experience

### Research Intern (Remote), Shenzhen Guangming Laboratory

Advisor: Prof. Wenhao Jiang

- *Revisiting Positional Priors in Vision Transformers with Semantic Gating (first author). Submitted to ICLR 2027.* Studies semantic gating beyond static geometry-driven bias in ViT attention.
- *Beyond Positional Encoding: Layerwise Attention Reconstruction via Query-Key Structural Priors (first author). Submitted to ICLR 2027.* Layerwise attention reconstruction using query-key structural priors.

### Graduate Research, University of Wisconsin–Madison

Advisor: Prof. Kangwook Lee

- *Basis-Mediated Bilinear Attention: A New Method for Greatly Reducing Query-Key Pathway Parameters. Under review (NeurIPS 2026 submission), current rating 4/4/4.* Proposes BMB/BMB-UV for query-key reparameterization in 7B-scale pretraining, reducing QK parameters by up to 95.83% while preserving model quality.

### Research Intern (Remote), Beihang Institute of Artificial Intelligence

2026

- *UnpairGR: Unpaired Modality-Agnostic Generative Recommendation. [PDF]* Proposes a modality-agnostic generative recommendation framework that learns a unified semantic ID space from unpaired multimodal observations, via item multimodal content embedding (MCE) and contrastive item representation learning (CIRL); achieves significant gains over ID-based and modality-based baselines on Amazon-book and Amazon-sports.

## Selected Talks

Graduate seminar talks, University of Wisconsin–Madison. Full slides on GitHub.

- *Newer Optimizers: SOAP, Muon, and MuonClip* 2025 [PDF]  
A unified view of preconditioned optimization: from Adagrad/Adam (diagonal) to Shampoo and SOAP (full-matrix curvature with Adam in the preconditioner's eigenbasis), then Muon's polar-factor update via Newton–Schulz iteration, why weight decay is required for Muon's weight-RMS equilibrium at scale, and QK-Clip for preventing attention logit explosion (as used in Kimi K2's 15.5T-token Muon pretraining).
- *Scaling Laws for Language Models* 2025 [PDF]  
Power-law relationships between loss and model size, data, and compute (Kaplan et al. 2020); Chinchilla's compute-optimal recipe and its three estimation methods (~20 tokens/param vs. Kaplan's 2); and inference-aware reallocation of the compute budget (Beyond Chinchilla).
- *SnapKV, SpecKV, and SpecPC: Efficient Long-Context Inference* Dec. 2025 [PDF]  
KV-cache compression via observation-window attention voting (SnapKV) and its hit-rate analysis; SpecKV's draft-model lookahead for importance estimation, with a worst-case error bound; and SpecPC for prompt compression, with consistent gains on RULER and LongBench.
- *Continual Adaptation via Prompt Evolution* Dec. 2025 [PDF]  
A survey of self-evolving agents formalized as recurring system transformations: what to evolve (context, tools, architecture), when (intra-/inter-test-time), how (reward-based, imitation, population-based), and where; plus PromptBreeder's self-referential mutation-prompt evolution, APE, and PromptAgent's MCTS planning for prompt optimization.
- *Language Diffusion Models* 2025 [PDF]

Two paths to diffusion language models: continuous latent-space diffusion (Diffusion-LM,  $\mathbf{x}_0$ -parametrization, clamping for word commitment) versus discrete masked diffusion (SEDD, LLaDA, Dream); the any-order autoregressive equivalence of mask-based diffusion, on-policy RL for efficient decoding, and connections between BERT/ENTP and one-step diffusion.

- *Verification and Reasoning-Focused Post-Training* Oct. 2025 [\[PDF\]](#)  
From GSM8K-style verifiers to outcome- vs. process-supervised reward models (PRM800K), generative verifiers (GenRM), and multi-domain VersaPRM; then reasoning-focused RL post-training: o1-style test-time compute scaling, DeepSeek-R1-Zero's emergent reflection, GRPO, and multi-stage RL pipelines.
- *Fantastic Pretraining Optimizers and Where to Find Them* (Stanford CS336 lecture slides) 2025 [\[PDF\]](#)

## Selected Projects

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### KI-VLA: $\pi 0.5$ Knowledge-Insulation Style VLA Training Framework

[GitHub](#) | [W&B](#)

- Implemented a fully tested minimal training framework reproducing the  $\pi 0.5$  / Knowledge-Insulation (KI) paradigm: the VLM backbone is trained with a unified next-token-prediction loss over discrete FAST action tokens, while a flow-matching action expert is co-trained with its gradients completely stop-gradient ("insulated") from the backbone; verified by a real-backprop insulation check plus a negative control.
- Supports two-stage training (VLM pre-training, then KI co-training) with pluggable backbones (Qwen3-VL / Cosmos-Reason2) and LeRobot-format robot data; inference via a single backbone forward pass plus a few Euler integration steps, with no autoregressive decoding.

### Global-Market-Newsboard

[GitHub](#)

- Developed an open-source market news dashboard focused on turning real-time information streams into a more usable interface for retail traders.
- Built a user-facing workflow spanning news aggregation, data handling, presentation logic, and practical decision-support design; deployed at [globalnewsboard.cn](https://globalnewsboard.cn) (demo password: 123456).

### Campus Lane Detection

[GitHub](#)

- Built a course project around ultra-fast deep lane detection in PyTorch using a structure-aware lane detection pipeline with training, evaluation, visualization, and speed-testing scripts.
- Adapted TuSimple / CULane-style tooling to campus driving footage and produced a 1080p / 30fps curved-lane overlay demo.

## Skills

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- **Distributed Pretraining:** PyTorch DDP, FSDP, ZeRO, DeepSpeed, Megatron-LM, mixed precision, gradient checkpointing
- **Memory-Efficient Attention:** FlashAttention, SDPA, MQA, GQA, MLA, KV-cache compression, quantization, activation checkpointing
- **Optimizers and Scaling:** AdamW, Lion, Shampoo, SOAP, Muon, QK-Clip; Chinchilla scaling laws, DataComp-LM, data mixture design
- **Post-Training:** SFT, RLHF, DPO-style preference optimization, LoRA, DoRA, QLoRA, instruction tuning, synthetic data refinement
- **Agents and Tool Use:** ReAct, Toolformer, DSPy, GEPA, Codex, Claude Code, Kimi, Swarm, multi-agent workflows

## Selected Coursework

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- Large Language Models and Deep Learning: [Advanced Topics in Large Language Models \(ECE 901\)](#); [Introduction to Artificial Neural Networks \(CS 539\)](#)
- Machine Learning and Optimization: [Machine Learning \(ECE 760\)](#); [Matrix Methods in Machine Learning \(ECE 532\)](#); [Topics in Mathematical Data Science \(ECE 888\)](#)
- Perception and Signal Processing: [Image Processing \(ECE 533\)](#)